INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
SALTLAKE
(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA

Term – II , Third Semester Examination, October-November 2026

Subject Code: BCACC301

Subject Name: Design and Analysis of Algorithms

Full Marks: 30 Set No:1 Duration: 1 Hour

Part - A
Answer any 5 questions out of 6
Each question carries 2 marks (2 × 5)

1. State two differences between the 0/1 Knapsack and the Fractional Knapsack problem. **2**
2. State the worst-case lower bound for comparison-based sorting. **2**
3. State the two key properties that typically indicate a problem is suitable for dynamic programming. **2**
4. Write the recurrence for the binomial coefficient C(n,k) used in dynamic programming. **2**
5. Define the Sum of Subsets problem. **2**
6. Define the class NP using polynomial-time verification. **2**

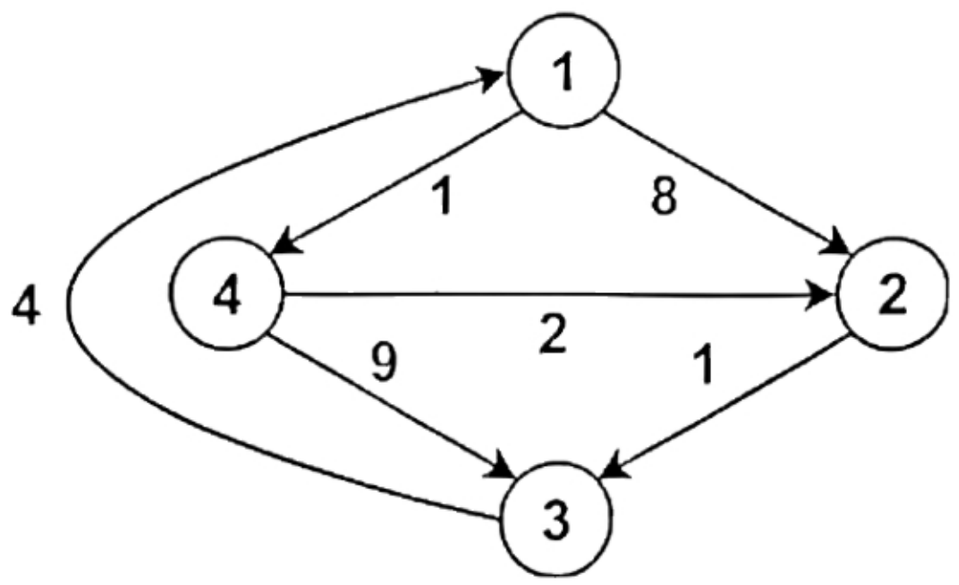
Part - B
Answer any 2 questions out of 3
Each question carries 5 marks (5 × 2)

7. Apply Prim's algorithm, starting from vertex A, to the undirected weighted graph with $V = \{A, B, C, D, E\}$ and $E = \{AB: 2, AC: 3, BC: 1, BD: 4, CD: 5, CE: 6, DE: 2\}$. Show the selected edges in order and calculate the total weight. **5**
8. Give a dynamic-programming solution to the 0-1 knapsack problem that runs in O(nW) time, where n is the number of items and W is the maximum weight of items that the thief can put in his knapsack. **5**
9. Differentiate P, NP, NP-hard and NP-complete. Give one standard example category for each where appropriate. **5**

Part - C
Answer any 1 question out of 2
Each question carries 10 marks (10 × 1)

10. Analyze Prim's and Kruskal's algorithms for MST in terms of time complexity and real-world suitability. **10**

11. Apply Floyd-Warshall to the weighted directed graph given bellow : **10**



Show matrices after each intermediate vertex and report all-pairs shortest distances.

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Full Marks: 30 Set No: 2 Duration: 1 Hour

Part - A

Answer any 5 questions out of 6

Each question carries 2 marks (2 × 5)

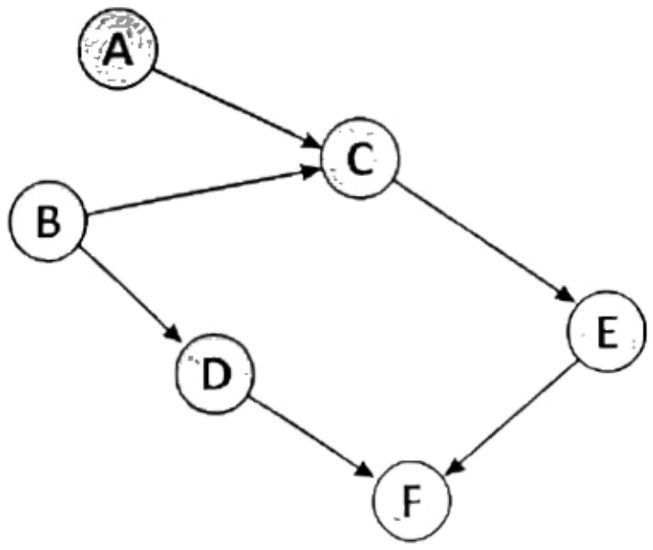
1. What is the main idea behind Strassen's matrix multiplication algorithm? **2**
2. State two differences between Prim's algorithm and Kruskal's algorithm. **2**
3. Differentiate memoization and tabulation method.
4. State the standard DP table dimensions and time complexity for 0/1 knapsack with n items and capacity W. **2**
5. Define a Hamiltonian circuit in a graph. **2**
6. What two conditions must be established to prove a problem NP-complete? **2**

Part - B

Answer any 2 questions out of 3

Each question carries 5 marks (5 × 2)

7. Explain the decision-tree model for comparison sorting. Illustrate it by sorting three distinct elements. **5**
8. Consider the following Directed Acyclic Graph with edges A-C, B-C, B-D, C-E, D-F, E-F, produce one valid topological order using indegree method. **5**



9. Define a Hamiltonian circuit. Explain its backtracking solution using a state-space tree and state the worst-case time complexity. **5**

Part - C
Answer any 1 question out of 2
Each question carries 10 marks (10 × 1)

10. Draw the major part of the state-space search for the 8-Queen problem and identify branches pruned because of column or diagonal conflicts. **10**

11. Solve 0/1 knapsack for capacity 10 with items A(Value,Weight) - A(10,5), B(40,4), C(30,6), D(50,3). Construct the complete Dynamic Programming table, identify the optimal value, and recover the selected items. **10**

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Full Marks: 30 Set No:3 Duration: 1 Hour

Part - A
Answer any 5 questions out of 6
Each question carries 2 marks (2 × 5)

1. State the recurrence relation and time complexity of Strassen's matrix multiplication algorithm. **2**
2. State the greedy criterion used by Prim's algorithm to select the next edge. **2**
3. What is the time complexity of computing the first n Fibonacci numbers by bottom-up dynamic programming? **2**
4. Can a DAG (Directed Acyclic Graph) have more than one valid topological ordering? Give a condition under which this happens. **2**
5. Define a vertex cover of an undirected graph. **2**
6. If any NP-complete problem is solved in polynomial time, what follows about P and NP? **2**

Part - B
Answer any 2 questions out of 3
Each question carries 5 marks (5 × 2)

7. Explain Dijkstra's algorithm with a proper example and state its time and space complexity. **5**
8. Consider the following Knapsack Problem, find an optimal solution using Dynamic Programming approach. **5**
   Weights: { 2, 3, 1, 4 }
   Profits: { 5, 6, 4, 8 }
   Knapsack capacity: 6
9. For the graph with edges AB, BC, CD, DA, AC, determine whether a Hamiltonian circuit exists and exhibit one if it does. **5**

Part - C

Answer any 1 question out of 2

Each question carries 10 marks (10 × 1)

10. Consider the graph with vertices {A, B, C, D, E, F, G} and edges AB, AC, BD, BE, CF, EG, FG, perform DFS from A using alphabetical adjacency order. Record discovery/finish times and classify the traversal edges appropriate to the graph type. **10**

11. Derive a dynamic programming algorithm for binomial coefficients. Compute C(8,4) and analyze time and space complexity, and show how the time can be reduced from $O(2^n)$ to O(nk). **10**

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